ACVP KAS IFC SSC Specification

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Abstract

This document defines the JSON schema for testing KAS-IFC-SSC SP800-56Br2 implementations with the ACVP specification.

Keywords

The following are keywords to be used by search engines and document catalogues.

ACVP; cryptography

Foreword

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Audience

This document is intended for the users and developers of ACVP.

Conventions

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 of [RFC 2119] and [RFC 8174] when, and only when, they appear in all capitals, as shown here.

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Executive Summary

The Automated Crypto Validation Protocol (ACVP) defines a mechanism to automatically verify the cryptographic implementation of a software or hardware crypto module. The ACVP specification defines how a crypto module communicates with an ACVP server, including crypto

capabilities negotiation, session management, authentication, vector processing and more. The ACVP specification does not define algorithm specific JSON constructs for performing the crypto validation. A series of ACVP sub-specifications define the constructs for testing individual crypto algorithms. Each sub-specification addresses a specific class of crypto algorithms. This sub-specification defines the JSON constructs for testing KAS-IFC-SSC SP800-56Br2 implementations using ACVP.

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1. Introduction

The Automated Crypto Validation Protocol (ACVP) defines a mechanism to automatically verify the cryptographic implementation of a software or hardware crypto module. The ACVP specification defines how a crypto module communicates with an ACVP server, including crypto capabilities negotiation, session management, authentication, vector processing and more. The ACVP specification does not define algorithm specific JSON constructs for performing the crypto validation. A series of ACVP sub-specifications define the constructs for testing individual crypto algorithms. Each sub-specification addresses a specific class of crypto algorithms. This sub-specification defines the JSON constructs for testing KAS-IFC-SSC SP800-56Br2 implementations using ACVP.

2. Supported KAS-IFC-SSC

The following shared secret computation and test revision pair **MAY** be advertised by the ACVP compliant cryptographic module:

- KAS-IFC-SSC / SP800-56Br2

3. Test Types and Test Coverage

The ACVP server performs a set of tests on the KAS protocol in order to assess the correctness and robustness of the implementation. A typical ACVP validation session **SHALL** require multiple tests to be performed for every supported permutation of KAS capabilities. This section describes the design of the tests used to validate implementations of KAS algorithms.

3.1. Test Types

There are two test types for KAS testing:

- "AFT" — Algorithm Function Test. In the AFT test mode, the IUT **SHALL** act as a party in the Key Agreement with the ACVP server. The server **SHALL** generate and provide all necessary information for the IUT to compute a shared secret z; both the server and IUT **MAY** act as party U/V.
- "VAL" — Validation test. In the VAL test mode, The ACVP server **MUST** generate a complete (from both party U and party V's perspectives) shared secret z, and expects the IUT to be able to determine if that shared secret is valid. Various types of conditions/errors **MUST** be introduced in varying portions of the key agreement process (changed Z, Z with leading zero nibble, etc), that the IUT **MUST** be able to detect and report on.

3.2. Test Coverage

The tests described in this document have the intention of ensuring an implementation is conformant to [SP 800-56B Rev. 2].

3.2.1. Requirements Covered

- SP 800-56Br2 — 5.3 Random Bit Generators. Though random values are used, the testing of the construction of those random values **SHALL NOT** be in scope of ACVP testing.
- SP 800-56Br2 — 6.2 Criteria for RSA Key Pairs for Key Establishment. The ACVP server **SHALL** support the three key generation methods of "basic", "prime factor", and "CRT".
- SP 800-56Br2 — 6.3 RSA Key-Pair Generators. The ACVP server **SHALL** utilize IUT provided RSA public keys, and generate key pairs to accommodate testing. The ACVP server **SHALL** support both fixed and random public exponents.
- SP 800-56Br2 — 7 Primitives and Operations. 7.1 Encryption and Decryption primitives as well as 7.2 Encryption and Decryption Operations **SHALL** be used for the calculation/ validation of the shared secret z.
- SP 800-56Br2 — 8 Key-Agreement Schemes. ACVP server testing will make use of "KAS1" and "KAS2" for testing of shared secret z. These exact schemes are not specified in the document, as "-basic" and "*-confirmation" make use of at least a KDF, if not also Key Confirmation. The intention of testing against the algorithm described in this document is specifically around the calculation of the shared secret z.

3.2.2. Requirements Not Covered

- SP 800-56Br2 — 5.3 Random Bit Generators. Though random values are used, the testing of the construction of those random values **SHALL NOT** be in scope of ACVP testing.
- KAS SSC testing only covers testing of SP800-56Br2 through the computation of a shared secret z. Additional functions of KAS as a whole such as KDF, KC, etc. **MAY** be covered within the scope of the full KAS testing; please see that document for further details.

4. Capabilities Registration

ACVP requires crypto modules to register their capabilities. This allows the crypto module to advertise support for specific algorithms, notifying the ACVP server which algorithms need test vectors generated for the validation process. This section describes the constructs for advertising support of KAS-IFC-SSC SP800-56Br2 algorithms to the ACVP server.

The algorithm capabilities **MUST** be advertised as JSON objects within the 'algorithms' value of the ACVP registration message. The 'algorithms' value is an array, where each array element is an individual JSON object defined in this section. The 'algorithms' value is part of the 'capability_exchange' element of the ACVP JSON registration message. See the ACVP specification [ACVP] for more details on the registration message.

4.1. Prerequisites

Each algorithm implementation **MAY** rely on other cryptographic primitives. For example, RSA Signature algorithms depend on an underlying hash function. Each of these underlying algorithm primitives must be validated, either separately or as part of the same submission. ACVP provides a mechanism for specifying the required prerequisites:

Prerequisites, if applicable, **MUST** be submitted in the registration as the `prereqVals` JSON property array inside each element of the `algorithms` array. Each element in the `prereqVals` array **MUST** contain the following properties

Table 1 — Prerequisite Properties

JSON Property	Description	JSON Type
algorithm	a prerequisite algorithm	string
valValue	algorithm validation number	string

A "valValue" of "same" **SHALL** be used to indicate that the prerequisite is being met by a different algorithm in the capability exchange in the same registration.

An example description of prerequisites within a single algorithm capability exchange looks like this

```
"prereqVals":
[
  {
    "algorithm": "Alg1",
    "valValue": "Val-1234"
  },
  {
    "algorithm": "Alg2",
    "valValue": "same"
  }
```

]

Figure 1

4.2. Prerequisite Algorithms

Some algorithm implementations rely on other cryptographic primitives. For example, IKEv2 uses an underlying SHA algorithm. Each of these underlying algorithm primitives must be validated, either separately or as part of the same submission. ACVP provides a mechanism for specifying the required prerequisites:

Table 2 — Prerequisite Algorithms JSON Values

JSON Value	Description	JSON Type	Valid Values	Optional
algorithm	a prerequisite algorithm	value	DRBG, RSA	No
valValue	algorithm validation number	value	actual number or “same”	No
prereqAlgVal	prerequisite algorithm validation	object with algorithm and valValue properties	see above	Yes

KAS has conditional prerequisite algorithms, depending on the capabilities registered:

Table 3 — Prerequisite requirement conditions

Prerequisite Algorithm	Condition
DRBG	Always **REQUIRED**
SHA	Always **REQUIRED**
RSA	RSA KeyGen validation **REQUIRED** when IUT makes use of the generation/validation of keys within the module boundary.

4.3. Algorithm Capabilities JSON Values

Each algorithm capability advertised is a self-contained JSON object using the following values.

Table 4 — KAS ECC Capabilities JSON Values

JSON Value	Description	JSON Type	Valid Values	Optional
algorithm	The algorithm under test	value	KAS-IFC-SSC	No
revision	The algorithm testing revision to use.	value	“Sp800-56Br2”	No

JSON Value	Description	JSON Type	Valid Values	Optional
prereqVals	Prerequisite algorithm validations	array of prereqAlgVal objects	See Section 4.2	No
keyGenerationMethods	The supported key generation methods.	array of string	See Section 4.4	No
modulo	The supported common modulo	array of integer	See Section 4.5	No
fixedPubExp	The fixed public exponent used for key generation. Required if using at least 1 static fixed public exponent key generation method.	hex		Yes
scheme	Array of supported key agreement schemes each having their own capabilities	object	See Section 4.6.1	No

Note: Some optional values are **REQUIRED** depending on the algorithm. Failure to provide these values will result in the ACVP server returning an error to the ACVP client during registration.

4.4. Supported Key Generation Methods

At least one key generation method is **REQUIRED** within the array. The following types **MAY** be advertised by the ACVP compliant crypto module:

- rsakpg1-basic — An RSA key pair with a private key in the basic format, and with a fixed public exponent.
- rsakpg1-prime-factor — An RSA key pair with a private key in the prime factor format, and with a fixed public exponent.
- rsakpg1-crt — An RSA key pair with a private key in the Chinese Remainder Theorem format, and with a fixed public exponent.
- rsakpg2-basic — An RSA key pair with a private key in the basic format, with a random public exponent.

- rsakpg2-prime-factor — An RSA key pair with a private key in the prime factor format, with a random public exponent.
- rsakpg2-crt — An RSA key pair with a private key in the Chinese Remainder Theorem format, with a random public exponent.

4.5. Supported Common Modulo

At least one supported common modulo is **REQUIRED** within the array. The following common modulo **MAY** be advertised by the ACVP compliant crypto module:

- 2048 — estimated security strength 112
- 3072 — estimated security strength 128
- 4096 — estimated security strength 152
- 6144 — estimated security strength 176
- 8192 — estimated security strength 200

4.5.1. Hash Function Z

An optional hash function that should be applied to z from both the ACVP server and IUT for comparison purposes. The provided `hashFunctionZ's security strength **MUST** be at least as strong as the greatest security strength modulo selected from Section 4.5

The following hash functions **MAY** be advertised by an ACVP compliant server:

- SHA-1
- SHA2-224
- SHA2-256
- SHA2-384
- SHA2-512
- SHA2-512/224
- SHA2-512/256
- SHA3-224
- SHA3-256
- SHA3-384
- SHA3-512

4.6. KAS IFC Schemes

All other scheme capabilities are advertised as a self-contained JSON object using the following values. Note that **AT LEAST** one valid scheme must be registered.

4.6.1. KAS IFC Scheme Capabilities JSON Values

KAS Schemes

- KAS1
- KAS2

Table 5 — KAS IFC Capabilities JSON Values

JSON Value	Description	JSON Type	Valid Values	Optional
kasRole	Roles supported for key agreement	array	initiator and/or responder	No

4.7. Example Registration

The following is a example JSON object advertising support for KAS IFC SSC.

```
{
  "vsId": 0,
  "algorithm": "KAS-IFC-SSC",
  "revision": "Sp800-56Br2",
  "scheme": {
    "KAS1": {
      "kasRole": [
        "initiator",
        "responder"
      ]
    },
    "KAS2": {
      "kasRole": [
        "initiator",
        "responder"
      ]
    }
  },
  "keyGenerationMethods": [
    "rsakpg2-basic",
    "rsakpg2-crt"
  ],
  "modulo": [
    2048
  ],
  "hashFunctionZ": "SHA2-512"
}
```

Figure 2

5. Generation Requirements per Party per Scheme

The various schemes of KAS/KTS all have their own requirements as to keys and nonces per scheme, per party. The below table demonstrates those generation requirements:

Table 6 — Required Party Generation Obligations

Scheme	KasMode	KasRole	KeyPair	Generates Cipher Text
KAS1	NoKdfNoKc	InitiatorPartyU	False	True
KAS1	NoKdfNoKc	ResponderPartyV	True	False
KAS2	NoKdfNoKc	InitiatorPartyU	True	True
KAS2	NoKdfNoKc	ResponderPartyV	True	True

6. Test Vectors

The ACVP server provides test vectors to the ACVP client, which are then processed and returned to the ACVP server for validation. A typical ACVP validation test session would require multiple test vector sets to be downloaded and processed by the ACVP client. Each test vector set represents an individual algorithm defined during the capability exchange. This section describes the JSON schema for a test vector set used with KAS-IFC-SSC SP800-56Br2 algorithms.

The test vector set JSON schema is a multi-level hierarchy that contains meta data for the entire vector set as well as individual test vectors to be processed by the ACVP client. The following table describes the JSON elements at the top level of the hierarchy.

Table 7 — Top Level Test Vector JSON Elements

JSON Values	Description	JSON Type
acvVersion	Protocol version identifier	string
vsId	Unique numeric vector set identifier	integer
algorithm	Algorithm defined in the capability exchange	string
mode	Mode defined in the capability exchange	string
revision	Protocol test revision selected	string
testGroups	Array of test groups containing test data, see Section 6.1	array

An example of this would look like this

```
{
  "acvVersion": "version",
  "vsId": 1,
  "algorithm": "Alg1",
  "mode": "Mode1",
  "revision": "Revision1.0",
  "testGroups": [ ... ]
}
```

Figure 3

6.1. Test Groups JSON Schema

The testGroups element at the top level in the test vector JSON object is an array of test groups. Test vectors are grouped into similar test cases to reduce the amount of data transmitted in the vector set. For instance, all test vectors that use the same key size would be grouped together. The Test Group JSON object contains meta data that applies to all test vectors within the group. The following table describes the secure hash JSON elements of the Test Group JSON object.

The test group for KAS/KTS IFC is as follows:

Table 8 — Vector Group JSON Object

JSON Value	Description	JSON Type	Optional
tgId	Numeric identifier for the test group, unique across the entire vector set.	value	No
testType	The type of test for the group (AFT or VAL).	value	No
scheme	The scheme in use for the group. See Section 4.6.1 for possible values.	value	No
kasRole	The group role from the perspective of the IUT.	value	No
keyGenerationMethod	The private key generation method for the group.	value	No
modulo	The modulo in use for key generation.	value	No
tests	The tests for the group.	Array of objects, See Section 6.2.	No

6.2. Test Case JSON Schema

Each test group contains an array of one or more test cases. Each test case is a JSON object that represents a single test vector to be processed by the ACVP client. The following table describes the JSON elements for each KAS/KTS IFC test vector.

Table 9 — Test Case JSON Object

JSON Value	Description	JSON Type	Optional
tcId	Numeric identifier for the test case, unique across the entire vector set.	value	No
serverN	RSA N value for the ACVP server's key.	value	Yes
serverE	RSA E value for the ACVP server's key.	value	Yes
serverP	RSA P value for the ACVP server's key.	value	Yes

JSON Value	Description	JSON Type	Optional
serverQ	RSA Q value for the ACVP server's key.	value	Yes
serverD	RSA D value for the ACVP server's key.	value	Yes
serverDmp1	RSA Dmp1 value for the ACVP server's key.	value	Yes
serverDmq1	RSA Dmq1 value for the ACVP server's key.	value	Yes
serverIqmp	RSA Iqmp value for the ACVP server's key.	value	Yes
iutN	RSA N value for the IUT's key.	value	Yes
iutE	RSA E value for the IUT's key.	value	Yes
iutP	RSA P value for the IUT's key.	value	Yes
iutQ	RSA Q value for the IUT's key.	value	Yes
iutD	RSA D value for the IUT's key.	value	Yes
iutDmp1	RSA Dmp1 value for the IUT's key.	value	Yes
iutDmq1	RSA Dmq1 value for the IUT's key.	value	Yes
iutIqmp	RSA Iqmp value for the IUT's key.	value	Yes
serverC	The ciphertext generated by the ACVP server, encrypted with the IUT's public key.	value	Yes
iutC	The ciphertext generated by the IUT, encrypted with the ACVP server's public key.	value	Yes
	The shared secret z.	value	Yes
hashZ	the shared secret z after being run through the registered hash function (when applicable).	value	Yes

6.3. Example Test Vectors JSON Object

The following is a example JSON object for test vectors sent from the ACVP server to the crypto module.

```
{
  "vsId": 0,
  "algorithm": "KAS-IFC-SSC",
```

```
 "mode": "",
 "revision": "Sp800-56Br2",
 "isSample": true,
 "testGroups": [
   {
     "tgId": 1,
     "testType": "AFT",
     "tests": [
       {
         "tcId": 1,
         "serverN":
"C649AC045BB1DBF46E4C03F2F89218D0431E5CAEF2E5287DC999A2D6AEA6378CBC537EB1C4D2F35EEC902AED
         "serverE": "0231E30EA607"
       },
       {
         "tcId": 2,
         "serverN":
"FC86CCE925571F96C86DD5DE5051CD5BA10EC97CF0951E674A1C44448E3E185F963B16BA366EA3BDAF6E48CA
         "serverE": "312C39FF83"
       },
       {
         "tcId": 3,
         "serverN":
"CBEA97A18EADA539F3383DCC9AD14D53F2347D0297BEBBF10D95D4A97846A3E86B1D23A227EC7F3DADD6F7BE
         "serverE": "028BB17DED"
       },
       {
         "tcId": 4,
         "serverN":
"A3145C5D1334185949F916B976A0F434CB0A964A4296259F4B67D70C014138EF23421A833409F002721158C0
         "serverE": "0198629BE7"
       },
       {
         "tcId": 5,
         "serverN":
"9C7102B454557D79BA2BBA2724B6CDBE1DF53270BBBA72FFB1BC41FB33278C7DBFB8CB7910CC27369C3F7704
         "serverE": "0530D5B083E1"
       }
     ],
     "scheme": "KAS1",
     "kasRole": "initiator",
     "keyGenerationMethod": "rsakpg2-basic",
     "modulo": 2048
   },
   {
     "tgId": 2,
```

```
      "testType": "AFT",
      "tests": [
        {
          "tcId": 6,
          "iutN":
"B1EC0D4310E9C1FDD3869552934A9A8AAF8D56481AD6DA9FB025E4411EE5D05F75BC7B8A59C84CF08A17A162
          "iutE": "0C4ACC1AB158D5",
          "iutP":
"BD1BFB5DE31BB6329BDCB5CF19D3EF092B1E030823AA5FEE7546C7535C9CD315EFF91A4F57EFC3F6BF9C349A
          "iutQ":
"F0DB0E710224CD6078B08F262914D3ADB75A327F4725DC0F0DA4E3C5658C4EEF843D6357AB048512551DB270
          "iutDmp1":
"0152F9A7453A909873978C1F8C80FEEC15E6E7794D2F2B51503BBEBB12B8E50D0F855FF826A836AC2A0D885B
          "iutDmq1":
"7088CE1732F6C77ACE5C5B83333017290B9FF202DD7207AFB11288D89D64E4924637027B1CB32E59B1E0B015
          "iutIqmp":
"7442F467D8ADE8E4E170B6836A1D520AE505AF84C3A74ECE0B7559E88091B0A0A6600011A5D065BC92A99A03
          "serverC":
"069891ACA1F6EBC1789DAC75696D347B0ADE86CCFA423A551736F2D79DB8194EDD91674392AD9E90E28ACFAB
        },
        {
          "tcId": 7,
          "iutN":
"9C737DC168F1F1C48F055EE0DEAF1B018AFDE9F1D3F79BD83066BACE6CF86B37E16ECD9688B0A9278DE4EB83
          "iutE": "3F6A2EE787",
          "iutP":
"D2569C32D3DDF20138536959DA14D1D6A50F37F6AB8EC409D0112552F548CD42E85BE5101DBFC6B89E764A61
          "iutQ":
"BE6A247849229A12F795ACDCC2A2E71CDA6A4EA8F37C7B0F2BB92558EE15896DBE29221AED41A0FB40C02B87
          "iutDmp1":
"C69C019BE0A4D0303238C812CB6D2081424BA282AADC3580AFD8CFB9FB8D29DE33E32A5F2CB7D7BC71FD8EF2
          "iutDmq1":
"60F45201E938A04CCB6235A64C14C533FBF6F59826C0189EA3A39C8279F5C12701AFA6D12DFCA9C9A8A7D7D7
          "iutIqmp":
"6E70C20E9FA4E6D6EE79B0AF9DD3E1DFAE4FD14CBB817D89421208E60E301EA60B9FAA4C1D72006988F36812
          "serverC":
"7FFBF0F3363435DD72DC5AE0560FFF63694A36D7D5431E92431B5F896750D85AF187BFA1812330C09DB06632
        },
        {
          "tcId": 8,
          "iutN":
"DC73CA09CBC5DF179D858DBE22076BA39B3D0653647C86476F5525DA3721650AD3C406CC701A1705E47F0FAE
          "iutE": "030BA3223807",
          "iutP":
"E680E1976429254F0F9159FD2069FCD3F1B7C2A9BA43137E46D7772502BFB8AC0A8A52DE5C3DF7D67331C059
```

```
        "iutQ":
"F4D64ABA36500B786E5CA91F2A4526753C5119B33CF6FD2E8BBA9301E39BFB9747867ED2FB89603748FC220B
        "iutDmp1":
"5DEB6D5833F48E28D62272F8F035BBB43B1C9B4B107C93C96406CB6ED36955D30B8D13DD6951AF76E55BF966
        "iutDmq1":
"657EFE3F0EFC609954394CD65AD414CF0F9B984C4932246195024CE8DE72B72F0F8F7C315549948E2D89551E
        "iutIqmp":
"8ABAB9429CC1701BB2C73C84F1D3687B5B26DA6695CE1714DE0D44FC5E0CE42F15FA40B2D790E1110F6F236E
        "serverC":
"C2EF3E526A11B413ABCB6DA03A1E0ECB882A1CB750327FA7FCC0C5196D5FD7AC50FE6A690F71DC67B6231F48
      },
      {
        "tcId": 9,
        "iutN":
"B43897D1D772D5A69C95F8E19B0A1AA6AFB05B084F458347E60E86092FD160174873AE20743C078842DCFD1D
        "iutE": "F7878E4B",
        "iutP":
"C6F76551221F1CE26E8F9F3724EB4269AF48CD73BDD1DF67D01156F05DED97A67D6770145D0B77C9C8379781
        "iutQ":
"E7E19F7DFF8F6F0FBFA30A37203D9D6609EA06B17D6FBEFD5036745BFBC2F32744CBF447918C0742C55AABBB
        "iutDmp1":
"B16178E54ECD8F8B7FD2188BD0CC088C127A9300C683E043BC4F058883C023B839510BB278E44AE4E5F1E913
        "iutDmq1":
"A87D125DED29B05035BA141CFF732C6EE40C0EC47FD1EC8C2DCF15427D833FE0ECE95439E852857A6FD1188F
        "iutIqmp":
"AEB85D2361C6B62F115E4BADE05E5C6E95B0B8B28B69D034FC74CFEE0FFF740BF7CB8D6480095E4CD37335DF
        "serverC":
"846AF208451055B128E45E8CA7A1021355E75261C0830DE1DE10CFBBC9A421B4D0690C1071B6346B3D907DAF
      },
      {
        "tcId": 10,
        "iutN":
"A4C62BAB1F1608417FC96DE24CEB9872823F5794B30FA1B2BDA7B3BC8B3F7791AC814EF5259151A55FB8BE5F
        "iutE": "3826704A0094CB",
        "iutP":
"BF1754F2A1643AEBA281D3816A3E745DC8323D115643439DE52FA5F3DE0BB6D942C58BFA7F717FFFF0A0439D
        "iutQ":
"DCBE64F1BE749E49F7EE1A9C11C5917A7ADC985384C3B9CFF1FD3B2AAA9718C9F37D5A84EA4DAFD4EC32C8B0
        "iutDmp1":
"B944CD247E11808AB70DF5D8CC989BE5EF16947B39E5547C53722B64CF88FB6725C1A641DB38FA76B9E3C4BE
        "iutDmq1":
"C64FEA3E7B6DFCA8448345729F15D952001240ACFA713300CD0399133B4281806BFCA4F229FC4A3112BE24CE
        "iutIqmp":
"8D542F8095B7D5242CC41320C35AE44347C309B1F8AA4944DE799891FF04E8335EBABC89454129B87B2638E5
        "serverC":
"3BA7202966866FB7160F765D62F9CA2DE6562CC3305ED980B0BAD2F0171839D3416178F96D126C406C0B85EE
```

```
        }
      ],
      "scheme": "KAS1",
      "kasRole": "responder",
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      "kasRole": "initiator",
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 "58AA75A09D7F391932A94E974A7A5C55B91A8F3537DBD4974CCA1D719A5CEB5F395EE89A3E421E9D01648FAF
          "hashZ":
 "3D7B6FAB0D642E2FF535F3A506AC9A6F43BABF01A223BF7C29FB6385D11DAE8E33BA0E46A7265D528F8E48A9
        }
      ],
      "scheme": "KAS2",
      "kasRole": "initiator",
      "keyGenerationMethod": "rsakpg2-basic",
      "modulo": 2048
    }
  ]
}
```


Figure 4

7. Test Vector Responses

After the ACVP client downloads and processes a vector set, it **MUST** send the response vectors back to the ACVP server. The following table describes the JSON object that represents a vector set response.

Table 10 — Vector Set Response JSON Object

JSON Value	Description	JSON type	Optional
acvVersion	Protocol version identifier	value	No
vsId	Unique numeric identifier for the vector set	value	No
testGroups	Array of JSON objects that represent each test vector group. See Table 11.	array	No

The testGroups section is used to organize the ACVP client response in a similar manner to how it receives vectors. Several algorithms **SHALL** require the client to send back group level properties in their response. This structure helps accommodate that.

Table 11 — Vector Set Group Response JSON Object

JSON Value	Description	JSON type	Optional
tgId	The test group Id	value	No
tests	Array of JSON objects that represent each test vector group. See Table 12.	array	No

The testCase section is used to organize the ACVP client response in a similar manner to how it receives vectors. Several algorithms **SHALL** require the client to send back group level properties in their response. This structure helps accommodate that.

Table 12 — Vector Set Test Case Response JSON Object

JSON Value	Description	JSON type	Optional
tcId	The test case Id	value	No
testPassed	Used in VAL test types, should the computation of a shared secret z been successful?	boolean	Yes
iutC	The ciphertext computed by the IUT (using the ACVP server's public key).	value	Yes
	shared secret z.	value	Yes

JSON Value	Description	JSON type	Optional
hashZ	The hash of the shared secret z for instances in which a hashFunctionZ is registered.	value	Yes

7.1. Example Test Results JSON Object

The following is an example JSON object for KAS-IFC test results sent from the crypto module to the ACVP server.


```
{
  "vsId": 0,
  "algorithm": "KAS-IFC-SSC",
  "mode": "",
  "revision": "Sp800-56Br2",
  "isSample": true,
  "testGroups": [
    {
      "tgId": 1,
      "tests": [
        {
          "tcId": 1,
          "iutC":
"BC52DE87B867C97ECA449712EEB9BDA2E806EF230FFDA78B2ADF09E75B1276D9243DDDB9C78C88D2E9F6DA70
          "hashZ":
"B48A0B2BADAC02C8018A8E6DCF5E3B999C816F137F8475F700B2951252968A272B738052BDBBF759B6DE7E25
        },
        {
          "tcId": 2,
          "iutC":
"3E84C8478A7EAF1E20A360427E8EC57CA81772AEC5153FF7348F38510366F7A07D1A09CFC49DB7F4BD4B1858
          "hashZ":
"473BC4F1CD3B42D4F2A77F54208CE3B9D6254E82D2AC68C806FD9219FFCA97CF1F8C8D0CB446AF6FBA1CAC06
        },
        {
          "tcId": 3,
          "iutC":
"9AEB3DD05F2A71018A7E25F14FF577F1E31538553C49A6EAC534CE835FCFCD4719937026C46F58ABA0F6A6E1
          "hashZ":
"8EE8AD2476627C57C0D994C14C5B11007B598D7FE9121970C507DBBA0C5F2555D49DB5FBB34802242DDF978B
        },
        {
          "tcId": 4,
          "iutC":
"56C9CDE42CD0792C043586456561AD2C2FA3224714DCB1A5DD80677B606890AD9DB7BF4D488FE19D1984A398
```

```
        "hashZ":
"96ED3D03DEA33E2BEAE8835CE7758308F035D28CEF2F9CEFF474FE8156878B516C045F345576DEE53F7582CA
      },
      {
        "tcId": 5,
        "iutC":
"26F490F8AEAADF2C39580CEDA771163FAD23476B1E33CE900DCB738E1C73DF38EF439A7D72E751A058FDA842
        "hashZ":
"BB0FCF7CDDBAC2A38DF7A7DDFD76324B5A8C4208846EFE50E5D2DC21472FD3D053D88A58979C4519EF58A2A5
      }
    ]
  },
  {
    "tgId": 2,
    "tests": [
      {
        "tcId": 6,
        "hashZ":
"72873353BA81E9E3F97C169180850C6A342FB70258AAE0CB9217CAE2736FCE48D66295BB776EF518A97A8649
      },
      {
        "tcId": 7,
        "hashZ":
"66456CCB689956E171E7BBA83FA7659A2DD313FEBD915F74D392195EF48DD250C702FB2004DAC9D2AAF787BA
      },
      {
        "tcId": 8,
        "hashZ":
"A62646DD4C19E41476E1FEC16CDAE7B9482EF54505A1FD111DBF4901D0D7E590B949B00C437A642F3BD8EF6C
      },
      {
        "tcId": 9,
        "hashZ":
"25053C4FFEBB4D5553E7AD9A8097A8A8BC17B2CFDF25F71888DAE69F77ABEDDFC01BF064E90D839B20BEF4DC
      },
      {
        "tcId": 10,
        "hashZ":
"F2369C65E6DEDFC2CEDFEC8DB2CE06488F6341B34E64941ED1A26276C4709526D758C8844733566D32711D8C
      }
    ]
  },
  {
    "tgId": 3,
    "tests": [
      {
```

```
        "tcId": 11,
        "iutC":
"7165F43606D75D9E0062C23BDAD81054AF1F5DD867E6DF01C63D91F172EAE9857570315DE6CA985C612624B5
        "hashZ":
"9DB8AE3B99C33BE2717443F204BB96A850212615310B035AC327F42558F37938244621FF4D9D3B95C8874B40
      },
      {
        "tcId": 12,
        "iutC":
"7717EDC81D539C9E75C679C7C7F25E4BD0FE88D1D0E8A1DD61E4FE35C34E7293D80A746102D1F250C8F4EE78
        "hashZ":
"280EDA51AA55D0CAB3D61251B0BE8AB1D9B89ADC8D0C53FEB1B3CC33A7662BF72F648F941AE8D6BD35DE143C
      },
      {
        "tcId": 13,
        "iutC":
"15334D04B783615DCD85751A26DC7CA999F748A21C524517A4E8AED82EAAC56205951388871F22178B278AF8
        "hashZ":
"DF71B0DB82C2E45F1B0DBDCCDA7A00995FAC3323008FC89775DD93A2BEE553F9B0558264876B05EB48E11B71
      },
      {
        "tcId": 14,
        "iutC":
"46BF8D84185254F526CA1BACB8EE2805245841D54E7434CE1CB3D578B1BAAC74273B46464233430F1CDCC1EA
        "hashZ":
"785E85B2F3693FDE598EBDA463D4A31BDCF6A8DD4122A1C625B361A6BA5654E322CEB863C3E6FB470687E143
      },
      {
        "tcId": 15,
        "iutC":
"97E01C6237F205CAE45CA143B872068B3E7CEC09F88277ED8253D336735B1E55D648F7F2E4C1EC5401B1B8EF
        "hashZ":
"B62B498E074E19911BF49D4C86DA2BC899550BFDC5880765F24ACB16731DB0F05AEC0563EB6D6F5B769368F9
      }
    ]
  },
  {
    "tgId": 4,
    "tests": [
      {
        "tcId": 16,
        "iutC":
"59918CE9E94E85F9B55761AEB56A9EC9316A4680F222014E61DE8E741F91927BAE90B598079D57620D018F6C
        "hashZ":
"5E925B5B8FBF09C85695B2DEE5F24C30AE5DD027A49A5DA3ACD880839D915BD0F591E56578598E947DF8F47F
      },
```

```
        {
          "tcId": 17,
          "iutC":
"2C1EF41FE21CD049F70E565FD1A583517DA56D8453EB49F7079DE5187F5991AE0A2E40CC5D66D532036908E2
          "hashZ":
"47D59912FCAC4E8DADD36631D30A37B33FF6EC033D71FAD6B81630B9A7FCAE22CD9EF627EC94683C577671D7
        },
        {
          "tcId": 18,
          "iutC":
"89B6E92E24812EE7E56E9DB0A710CCD12A92B2DCC30521980F87548E45A9F08A9A55E60B641394D7605180DD
          "hashZ":
"F37AE64EFC034F023335E05ED40D2578506AC97963A68E17D8964713A8D5339DBEC08A54271C72BA9B7966B3
        },
        {
          "tcId": 19,
          "iutC":
"AD8870E7FCE18E54127FD3C07B02D71BDD10E519C414320E1D8E61420BF1907990E7C12A9B6003B2E64F2A38
          "hashZ":
"0747F513741CE872489521180B33A569F60F3E3CBF8102A49EBECAA3AA309AD7B8463543ADB867BC80973C34
        },
        {
          "tcId": 20,
          "iutC":
"089EEE5C8B64FDEE6C4B79EE0CC903321A0B2F14CE3A89820E38FC2A34B666CE0A3B41C41E5F7E1975F47DBA
          "hashZ":
"1487575F667890C8693BC8A731F55590A5E94FA9171E1A324EAC37092B049E9FB7506B09D4790C7C3397BCC3
        }
      ]
    },
    {
      "tgId": 5,
      "tests": [
        {
          "tcId": 21,
          "testPassed": false
        },
        {
          "tcId": 22,
          "testPassed": true
        },
        {
          "tcId": 23,
          "testPassed": true
        },
        {
```

```
        "tcId": 24,
        "testPassed": true
      },
      {
        "tcId": 25,
        "testPassed": true
      }
    ]
  },
  {
    "tgId": 6,
    "tests": [
      {
        "tcId": 26,
        "testPassed": true
      },
      {
        "tcId": 27,
        "testPassed": true
      },
      {
        "tcId": 28,
        "testPassed": true
      },
      {
        "tcId": 29,
        "testPassed": false
      },
      {
        "tcId": 30,
        "testPassed": true
      }
    ]
  },
  {
    "tgId": 7,
    "tests": [
      {
        "tcId": 31,
        "testPassed": true
      },
      {
        "tcId": 32,
        "testPassed": true
      },
      {
```

```
          "tcId": 33,
          "testPassed": true
        },
        {
          "tcId": 34,
          "testPassed": false
        },
        {
          "tcId": 35,
          "testPassed": true
        }
      ]
    },
    {
      "tgId": 8,
      "tests": [
        {
          "tcId": 36,
          "testPassed": true
        },
        {
          "tcId": 37,
          "testPassed": true
        },
        {
          "tcId": 38,
          "testPassed": true
        },
        {
          "tcId": 39,
          "testPassed": false
        },
        {
          "tcId": 40,
          "testPassed": true
        }
      ]
    }
  ]
}
```


Figure 5

8. Security Considerations

There are no additional security considerations outside of those outlined in the ACVP document.

Appendix A — Terminology

For the purposes of this document, the following terms and definitions apply.

A.1.

Prompt
JSON sent from the server to the client describing the tests the client performs
Registration
The initial request from the client to the server describing the capabilities of one or several algorithm, mode and revision combinations
Response
JSON sent from the client to the server in response to the prompt
Test Case
An individual unit of work within a prompt or response
Test Group
A collection of test cases that share similar properties within a prompt or response
Test Vector Set
A collection of test groups under a specific algorithm, mode, and revision
Validation
JSON sent from the server to the client that specifies the correctness of the response

Appendix B — Abbreviations and Acronyms

ACVP	Automated Crypto Validation Protocol
JSON	Javascript Object Notation

Appendix C — Revision History

Table C-1

Version	Release Date	Updates
1	2020-08-19	Initial Release

Appendix D — References

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Unresolved directive in src/kas/sp800-56br2/ssc/sections/98-references.adoc — include::../../common/common-sections/99-references-acvp.adoc[]